Lecture 9 Exercise Solutions

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1 Important Equations

Linear Regression

The estimated best line is given as:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_x x \tag{1}$$

Where the best fit is calculated using least squares as:

$$\hat{\beta}_1 = \frac{\sum_i^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_i^n (x_i - \bar{x})^2} = \frac{S_{xY}}{S_{xx}} \tag{2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x} \tag{3}$$

The residual is then given as:

$$\hat{\varepsilon}_i = Y_i - \hat{Y}_i \tag{4}$$

Correlation Coefficient

The correlation coefficient for the linear fit is given as:

$$R^2 = \frac{\sum_i^n (\hat{Y}_i - \bar{Y})^2}{\sum_i^n (Y_i - \bar{Y})^2} = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T}, \quad 0 \leq R^2 \leq 1 \tag{5}$$

Where a value closer to 1 means a better fit.

Variance Estimate

The variance is estimated as:

$$\hat{\sigma}^2 = \frac{SS_E}{n-2} \tag{6}$$

Where:

$$SS_E = SS_T - \hat{\beta}_1 S_{xY}, \qquad SS_T = \sum_i^n (Y_i - \bar{Y})^2 = \sum_i^n Y_i^2 - n\bar{Y}^2$$

Test For Significance of Regression

We can test for the significance of regression with:

$$H_0 : \beta_1 = 0, \quad H_1 : \beta_1 \neq 0$$

Which means that if we cannot reject H_0 then there is no linear relationship. This has the test statistic:

$$T_0 = \frac{\hat{\beta}_1 - \hat{\beta}_{1,0}}{\sqrt{\hat{\sigma}^2 / S_{xx}}} \tag{7}$$

2 Exercise 11.2.1

Diabetes and obesity are serious health concerns in the United States are much of the developed world. Measuring the amount of body fat a person carries is one way to monitor weight control progress, but measuring it accurately involves either expensive X-ray equipment or a pool in which to dunk the subject. Instead body mass index (BMI) is often used as a proxy for body fat because it is easy to measure.

$$\text{BMI} = \frac{\text{mass (kg)}}{(\text{height (m)})^2} = 703 \frac{\text{mass (lb)}}{(\text{height (in)})^2}$$

In a study of 250 men af Brigham Young University, both BMI and body fat were measured. Researchers found the following summary of statistics:

$$\sum_i^n x_i = 6322.28, \qquad \sum_i^n x_i^2 = 162674.18,$$
$$\sum_i^n y_i = 4757.90, \qquad \sum_i^n y_i^2 = 107679.27,$$
$$\sum_i^n x_i y_i = 125471.10$$

a. Calculate the least squares estimate of the slope and intercept. Graph the regression line.

First we calculate the means:

$$\bar{x} = \frac{\sum_i^n x_i}{n} = \frac{6322.28}{250} = 25.2891$$

$$\bar{Y} = \frac{\sum_i^n Y_i}{n} = \frac{4757.9}{250} = 19.0316$$

In order to find the least squares estimate we use eq. 2 & 3 and the information provided. From eq. 11.10 & 11.11 in the book:

$$S_{xx} = \sum_i^n x_i^2 - \frac{\left(\sum_i^n x_i\right)^2}{n} = 162674.18 - \frac{6322.28^2}{250} = 2789.3$$

$$S_{xY} = \sum_i^n x_i Y_i - \frac{\left(\sum_i^n x_i\right)\left(\sum_i^n Y_i\right)}{n} = 125471.1 - \frac{6322.28 \cdot 4757.9}{250} = 5148$$

Meaning:

$$\hat{\beta}_1 = \frac{S_{xY}}{S_{xx}} = \frac{5148}{2789.3} = 1.8456$$

Now we calculate:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x} = 19.0316 - 1.8456 \cdot 25.2891 = -27.64$$

The least squares fit is then:

$$\hat{Y} = 1.8456x - 27.64$$

Which is plotted in figure 1.

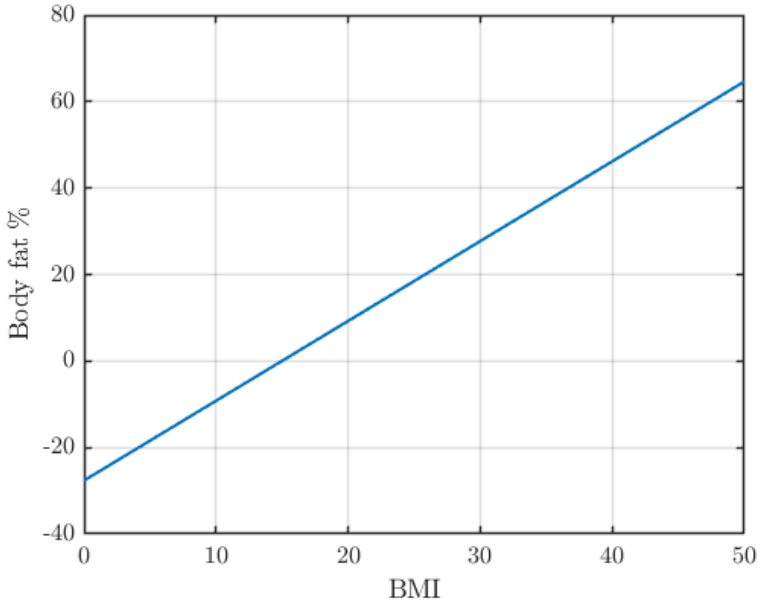


Figure 1: Linear regression of the body fat % as a function of the BMI.

b. Use the equation of the fitted line to predict what body fat would be observed, on average, for a man with a BMI of 30.

We insert $x = 30$:

$$\hat{Y}(30) = 1.8456 \cdot 30 - 27.64 = 27.73$$

c. Suppose that the observed body fat of a man with a BMI of 25 is 25%. Find the residual for that observation.

First the amount of body fat is estimated for a BMI of 25:

$$\hat{Y}(25) = 1.8456 \cdot 25 - 27.64 = 18.5$$

The residual is then calculated from eq. 4:

$$\hat{\varepsilon} = 25 - 18.5 = 6.5$$

d. Was the prediction for the BMI of 25 in part (c.) an overestimate or an underestimate? Explain briefly.

Because the residual is defiend as:

$$\hat{\varepsilon} = Y_i - \hat{Y}_i$$

And the residual was positive that means that $\hat{Y}$ underestimated the body fat % for the BMI of 25 in c.

3 Exercise 11.2.3

An article in *Concrete Research* presented data on compressive strength x and instrinsic permeability y of various concrete mixes and cures. Summary quantities are $n = 14$,

$$\sum_i^n x_i = 43, \qquad \sum_i^n x_i^2 = 157.42,$$
$$\sum_i^n y_i = 572, \qquad \sum_i^n y_i^2 = 23530,$$
$$\sum_i^n x_i y_i = 1697.80$$

a. Calculate the least squares estimates of the slope and intercept. Estimate σ^2. Graph the regression line.

The means are:

$$\bar{x} = \frac{\sum_i^n x_i}{n} = \frac{43}{14} = 3.07$$
$$\bar{Y} = \frac{\sum_i^n Y_i}{n} = \frac{572}{14} = 40.86$$

Using the same approach as in 11.2.1:

$$S_{xx} = \sum_i^n x_i^2 - \frac{\left(\sum_i^n x_i\right)^2}{n} = 157.42 - \frac{43^2}{14} = 25.35$$

$$S_{xY} = \sum_i^n x_i Y_i - \frac{\left(\sum_i^n x_i\right)\left(\sum_i^n Y_i\right)}{n} = 1697.80 - \frac{43 \cdot 572}{14} = -59.06$$

Meaning:

$$\hat{\beta}_1 = \frac{S_{xY}}{S_{xx}} = \frac{-59.06}{1525.3} = -2.33$$
$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x} = 40.86 - (-0.0387) \cdot 3.07 = 48.01$$

The fit is then:

$$\hat{Y} = -2.33x + 48.01$$

Which is plotted in figure 2.

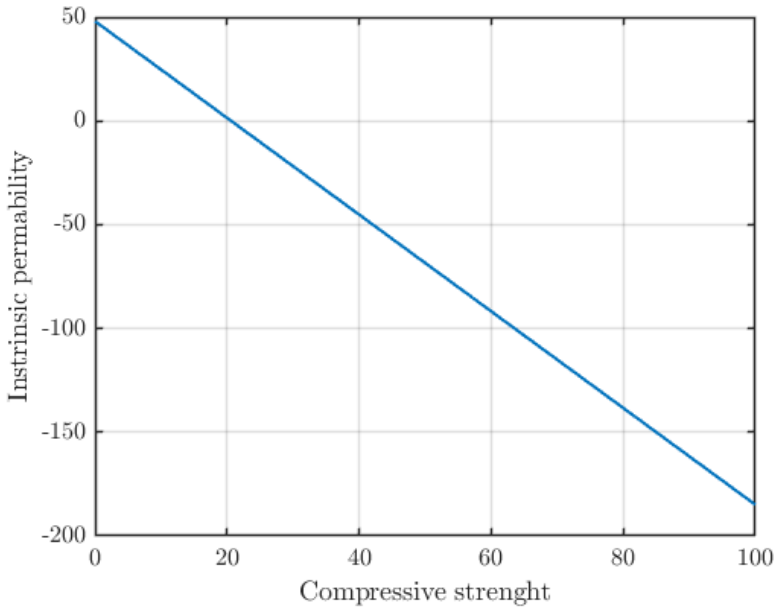


Figure 2: Linear regression of the intrinsic permeability as a function of the compressive strength of concrete.

The variance is estimated using eq. 6:

$$SS_T = \sum_i^n Y_i^2 - n\bar{Y}^2 = 23530 - 14 \cdot 40.86^2 = 159.7$$

$$SS_E = SS_T - \hat{\beta}_1 S_{xY} = 159.7 - (-2.33) \cdot (-59.06) = 22.12$$

The estimated variance is then:

$$\hat{\sigma}^2 = \frac{SS_E}{n-2} = \frac{22.12}{14-2} = 1.844$$

b. Use the equation of the fitted line to predict what the permeability would be observed when the compressive strength is $x = 4.3$

We insert into the equation for the fit:

$$\hat{Y}(4.3) = -2.33 \cdot 4.3 + 48.01 = 38$$

c. Give a point estimate of the mean permeability when the compressive strength is $x = 3.7$

We again use the equation for the fit:

$$\hat{Y}(3.7) = -2.33 \cdot 3.7 + 48.01 = 39.39$$

d. Suppose that the observed value of the permeability at $x = 3.7$ **is** $y = 46.1$**. Calculate the value of the corresponding residual.**

The esidual is calculated as:

$$\hat{\varepsilon}(3.7) = Y(3.7) - \hat{Y}(3.7) = 46.1 - 39.39 = 6.71$$

4 Exercise 11.2.6

An article in the *Journal of Environmental Engineering* reported the results of a study on the occurrence of sodium and chloride in surface streams in central Rhode Island. The following data are chloride concentration y (in milligrams per liter) and roadway area in the watershed x (in percentage).

y	4.4	6.6	9.7	10.6	10.8	10.9
x	0.19	0.15	0.57	0.70	0.67	0.63
y	11.8	12.1	14.3	14.7	15.0	17.3
x	0.47	0.70	0.60	0.78	0.81	0.78
y	19.2	23.1	27.4	27.7	31.8	39.5
x	0.69	1.30	1.05	1.06	1.74	1.62

a. Draw a scatter diagram of the data. Does a simple linear regression model seem appropriate here?

Data in MATLAB format:

x = [0.19 0.15 0.57 0.70 0.67 0.63 0.47 0.70 0.60 0.78 0.81 0.78 0.69 1.30 1.05 1.06 1.74 1.62];

y = [4.4 6.6 9.7 10.6 10.8 10.9 11.8 12.1 14.3 14.7 15.0 17.3 19.2 23.1 27.4 27.7 31.8 39.5];

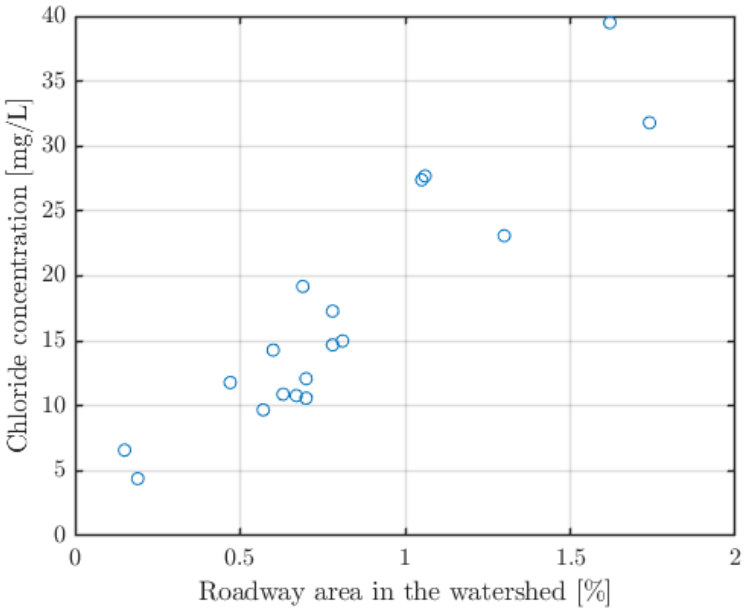


Figure 3: Scatter plot of data.

The scatter plot of the date (fig. 3) shows a linear trend so a linear regression model is appropriate.

b. Fit the simple the simple linear regression model using the method of least squares. Find an estimate of σ^2.

To find the regression model we first calculate the means with $n = 18$:

$$\bar{x} = \frac{1}{n}\sum_{i}^{n} x_i = 0.8061$$

$$\bar{y} = \frac{1}{n}\sum_{i}^{n} y_i = 17.05$$

Now as before:

$$S_{xx} = \sum_{i}^{n}(x_i - \bar{x})^2 = 3.0106$$

$$S_{xy} = \sum_{i}^{n}(x_i - \bar{x})(y_i - \bar{y}) = 61.9205$$

The fit can now be estimated:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{61.9205}{3.0106} = 20.57$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 17.05 - 2057 \cdot 0.8061 = 0.47$$

The linear regression model is then:

$$\hat{y} = 20.57x - 0.47$$

Which is plotted with the data in fig. 4

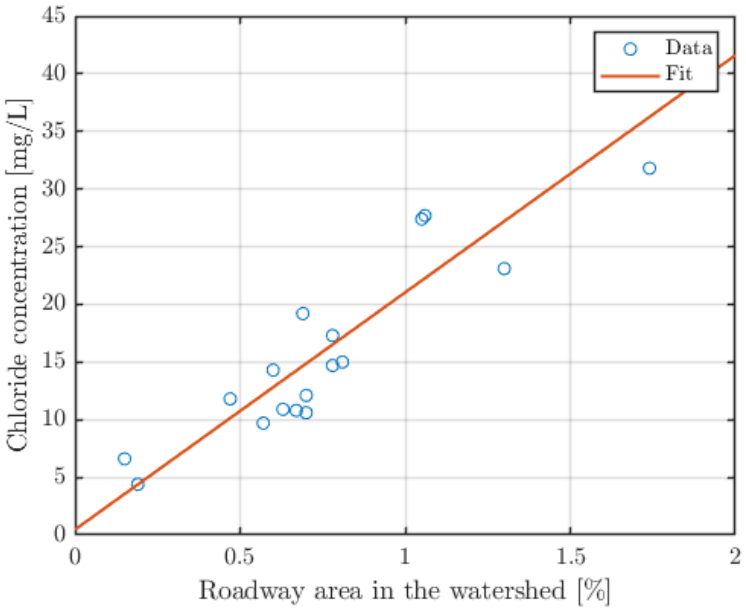


Figure 4: Scatter plot of data along fit linear fit.

the variance can now be estimated as before:

$$SS_T = \sum_i^n y_i^2 - n\bar{y}^2 = 1494.5$$

$$SS_E = SS_T - \hat{\beta}_1 S_{xy} = 1494.5 - 20.57 \cdot 61.9205 = 220.9472$$

The estimated variance is then:

$$\hat{\sigma}^2 = \frac{SS_E}{n-2} = \frac{220.9472}{18-2} = 13.81$$

c. Estimate the mean chloride concentration for a watershed that has 1% roadway area.

We use the model found in b. to calculate the estimated mean chloride concentration with $x = 1\%$:

$$\hat{y}(1\%) = \hat{\beta}_1 x + \hat{\beta}_0 = 20.57 \cdot 1\% - 0.47 = 21.04 \,\frac{\text{mg}}{\text{L}}$$

d. Find the fitted value corresponding to $x = 0.47$ and the associated residual.

The estimated value is:

$$\hat{y}(0.47\%) = \hat{\beta}_1 x + \hat{\beta}_0 = 20.57 \cdot 0.47\% - 0.47 = 10.14 \,\frac{\text{mg}}{\text{L}}$$

And from the dataset it can be seen that $y(0.47\%) = 11.8\,\text{mg/L}$ meaning the residual is:

$$\hat{\varepsilon}(0.47\%) = y(0.47\%) - \hat{y}(0.47\%) = 11.8 \,\frac{\text{mg}}{\text{L}} - 10.14 \,\frac{\text{mg}}{\text{L}} = 1.66 \,\frac{\text{mg}}{\text{L}}$$

5 Exercise 11.2.10

Show that in a simple linear regression model the point $(\bar{x}, \bar{y})$ lies exactly on the least squares regression line.

It is given that:

$$\hat{y} = \hat{\beta}_1 x + \hat{\beta}_0 \Rightarrow \hat{\beta}_0 = \hat{y} - \hat{\beta}_1 x$$

If we insert $(\bar{x}, \bar{y})$:

$$\hat{y} = \hat{\beta}_1 \bar{x} + \left(\bar{y} - \hat{\beta}_1 \bar{x}\right) = \bar{y}$$

We see that $(\bar{x}, \bar{y})$ lies on the regression line.

6 Exercise 11.7.3

Refer to the compressive strength data in Exercise 11.2.3. Use the summary statistics provided to calculate R^2 and provide a practical interpretation of this quantity.

To calculate R^2 we use eq. 5. Since we calculated SS_T and SS_E in 11.2.3 then:

$$R^2 = 1 - \frac{22.12}{159.7} = 0.8615$$

Which means that the linear regression model explains 86.15% of the total variation in the data.

7 Exercise 11.7.5

Refer to exercise 11.2.6 which presented data on chloride concentration y and roadway area x.

a. What proportion of the total variability in chloride concentration is accounted for by the regression model?

We can find the total variability accounted for by the model with an R^2 value using eq. 5. Since we calculated SS_T and SS_E in the exercise then:

$$R^2 = 1 - \frac{SS_E}{SS_T} = 1 - \frac{220.9472}{1494.5} = 0.8522$$

meaning 85.22% of the total variability is explained by the model.

b. Plot the residuals versus $\hat{y}$ and versus x. Interpret these plots. The residuals is calculated as:

$$\hat{\varepsilon}_i = y_i - \hat{y}_i$$

This is plotted as a function of x and $\hat{y}$ in figure 5 using the "stem(x,y)" command in MATLAB. In the linear regression model there were two assumptions:

$$E[\hat{\varepsilon}] = 0, \qquad V[\hat{\varepsilon}_i] = \text{constant} = \sigma^2$$

This means these plots should have random sign and amplitude throughout the plot without showing any trends fx. growing amplitude of residuals as a function of $\hat{y}$.

From the plots it looks reasonable to make the stated assumptions and use the least squares linear regression model.

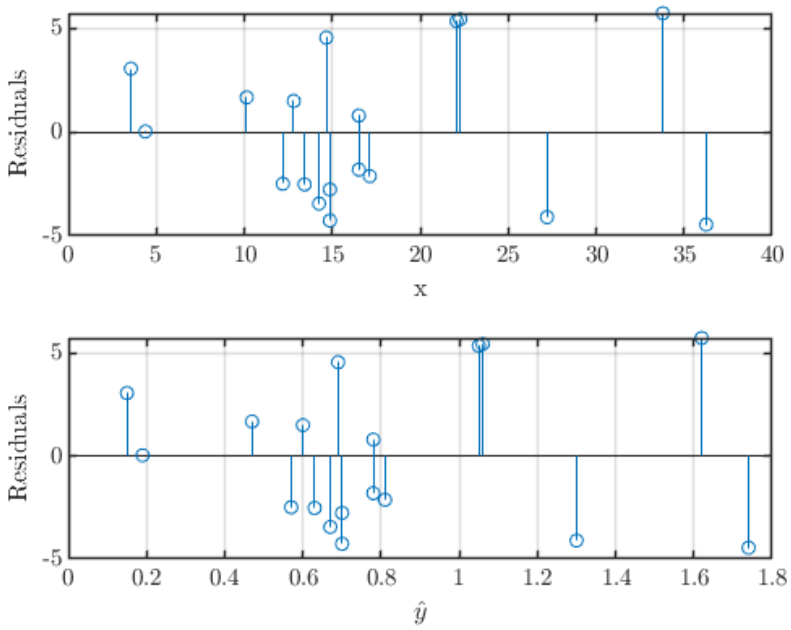


Figure 5: Plot of residuals as a function of x and $\hat{y}$

c. Prepare a normal probability plot of the residuals. Does the normality assumption appear to be satisfied?

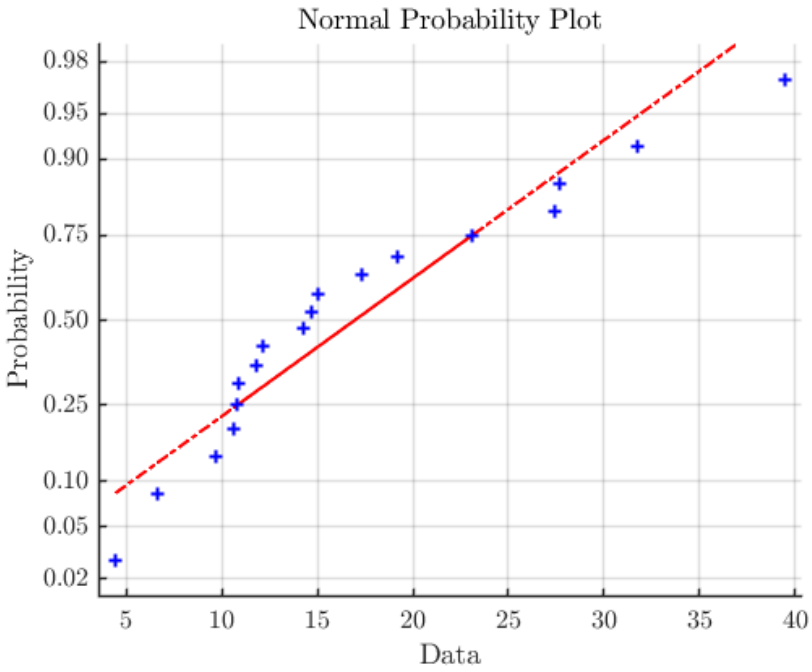


Figure 6: Normal probability plot.

From figure 6 it appears that there could be some systematic deviation from the line. This is seen from the curved appearance of the points. So a normal assumption is questionable.

8 Exercise 11.9.1

Determine if the following models are intrinsically linear. If yes, determine the appropriate transformation to generate the linear model.

a. $Y = \beta_0 x^{\beta_1} \epsilon$**.**

We can make a logarithmic transformation $Y \to Y^\star$:

$$Y^\star = \ln(Y) = \ln(\beta_0 x^{\beta_1} \epsilon) = \ln(\beta_0) + \beta_1 \ln(x) + \ln(\epsilon) = \ln(\beta_0) + \beta_1 x^\star + \ln(\epsilon)$$

Which turns out to be a straight line from the logarithmic transformation.

b. $Y = \frac{3+5x}{x} + \epsilon$

There is no parameters to estimate in this model so it is not a linear regression model.

c. $Y = \beta_0 \beta_1^x \epsilon$

We can make a logarithmic transformation:

$$Y^\star = \ln(Y) = \ln(\beta_0 \beta_1^x \epsilon) = \ln(\beta_0) + \ln(\beta_1)x + \ln(\epsilon)$$

d. $Y = \frac{x}{\beta_0 x + \beta_1 + x\epsilon}$

By taking the inverse we can get a transfomration to get a linear model:

$$Y^\star = Y^{-1} = \frac{\beta_0 x + \beta_1 + x\epsilon}{x} = \beta_0 + \beta_1 \frac{1}{x} + \epsilon = \beta_0 + \beta_1 x^\star + \epsilon$$

9 Exercise 11.9.2

The vapor pressure of water at various temperatures is in the table below:

Observation number	Temperature (K)	Vapor pressure (mmHg)
1	273	4.6
2	283	9.2
3	293	17.5
4	303	31.8
5	313	55.3
6	323	92.5
7	333	149.4
8	343	233.7
9	353	355.1
10	363	525.8
11	373	760.0

a. Draw a scatter diagram of these data. What type of relationship seems appropriate in relating y to x?

The data in MATLAB format: T = [273 283 293 303 313 323 333 343 353 363 373];
p = [4.6 9.2 17.5 31.8 55.3 92.5 149.4 233.7 355.1 525.8 760.0];

From fig. 7 an exponential relationship seems appropriate for pressure as a function of temperature.

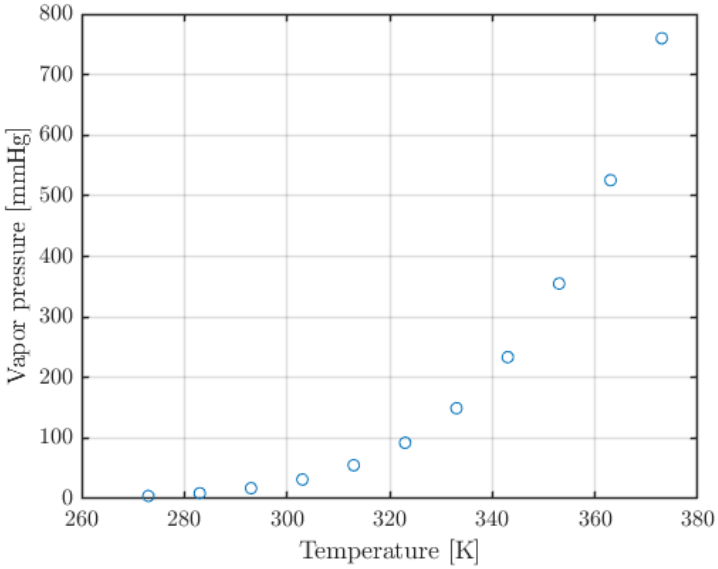


Figure 7: Scatter plot of data.

b. Fit a simple linear regression model to these data.

We can as before make a linear regression model. First we find the mean with $n = 11$:

$$\bar{T} = \frac{1}{n}\sum_{i}^{n} T_i = 323\,\mathrm{K}$$

$$\bar{p} = \frac{1}{n}\sum_{i}^{n} p_i = 203.1727\,\mathrm{mmHg}$$

With this we calculate:

$$S_{xx} = \sum_{i}^{n}(T_i - \bar{T})^2 = 11000$$

$$S_{xy} = \sum_{i}^{n}(T_i - \bar{T})(p_i - \bar{p}) = 73541$$

We can now estimate the coefficients:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{73541}{11000} = 6.6855$$

$$\hat{\beta}_0 = \bar{p} - \hat{\beta}_1\bar{T} = 203.1727 - 6.6855 \cdot 323 = -1956.3$$

meanign the linear regression model is:

$$\hat{p} = 6.6855T - 1956.3$$

This model is plotted with the data in figure. 8.

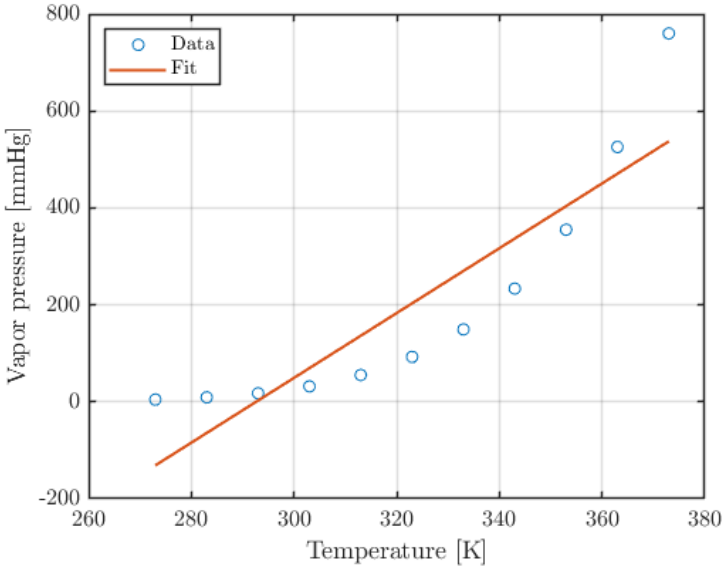


Figure 8: Scatter plot of data with linear fit.

c. Test for significance of regression using $\alpha = 0.05$. What conclusions can you draw?

We can test the signficance of regression using the test statistic in eq. 7 with $\hat{\beta}_{1,0} = 0$. First we need to estimate the variance where we first calculate

$$SS_T = \sum_i^n p_i^2 - n\bar{p}^2 = 616060$$

$$SS_E = SS_T - \hat{\beta}_1 S_{xy} = 124400$$

The estimated variance is then:

$$\hat{\sigma}^2 = \frac{SS_E}{n-2} = \frac{124400}{11-2} = 13823$$

The test statistic is then:

$$t_0 = \frac{\hat{\beta}_1 - \hat{\beta}_{1,0}}{\sqrt{\hat{\sigma}^2/S_{xx}}} = \frac{6.6855 - 0}{\sqrt{13823/11000}} = 5.964$$

We can now compare with:

$$t_{n-1,\frac{\alpha}{2}} = \text{tinv}\left(1 - \frac{\alpha}{2}, n-1\right) = \text{tinv}\left(1 - \frac{0.05}{2}, 11-1\right) = 2.2281$$

Because $t_0 > t_{n-1,\frac{\alpha}{2}}$ we can reject H_0 i.e. there is not evidence to reject a linear relationship.

d. Plot the residuals from the simple linear regression model versus $\hat{y}_i$. What do you conclude about model adequacy?

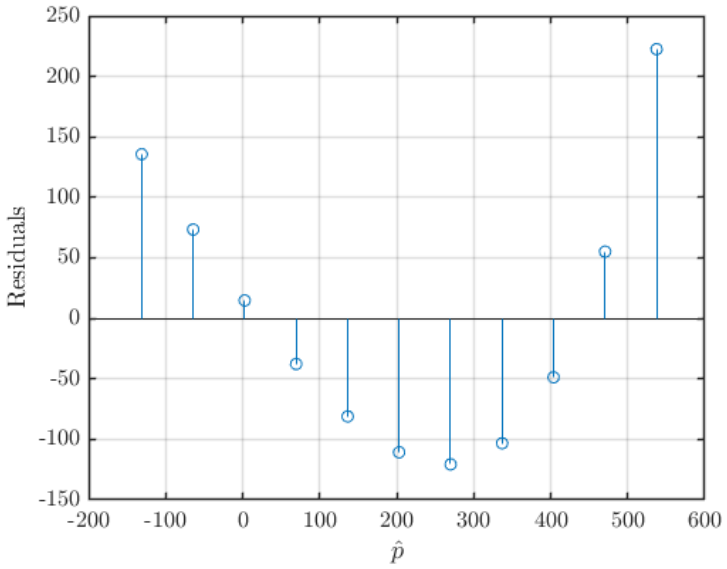


Figure 9: Plot of residuals.

From figure 9 it can be seen that the residuals have some systematic behavior which comes from the data looking exponential and not linear.

e. The Clausius-Clapeyron relationship states that

$$\ln(P_v) \propto -\frac{1}{T}$$

where P_v is the vapor-pressure of water. Repeat parts (a)-(d) using an appropiate transformation.

We use the Clausius-Clapeyron relationship and transform so $p^\star = \ln(p)$ and $T^\star = 1/T$ seen in figure 10.

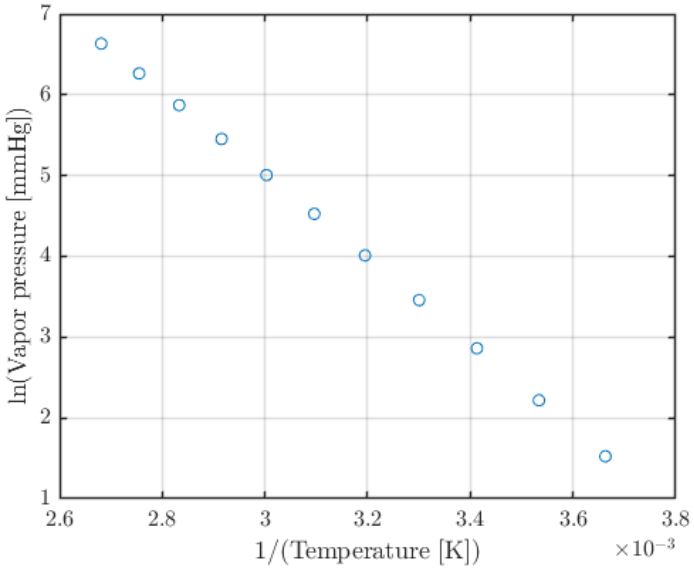


Figure 10: Plot of transformed data.

Now we can find the regression coefficients again as:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = -5200.8$$

$$\hat{\beta}_0 = 20.61$$

Meaning the model is:

$$p^\star = -5200.8T^\star + 20.61$$

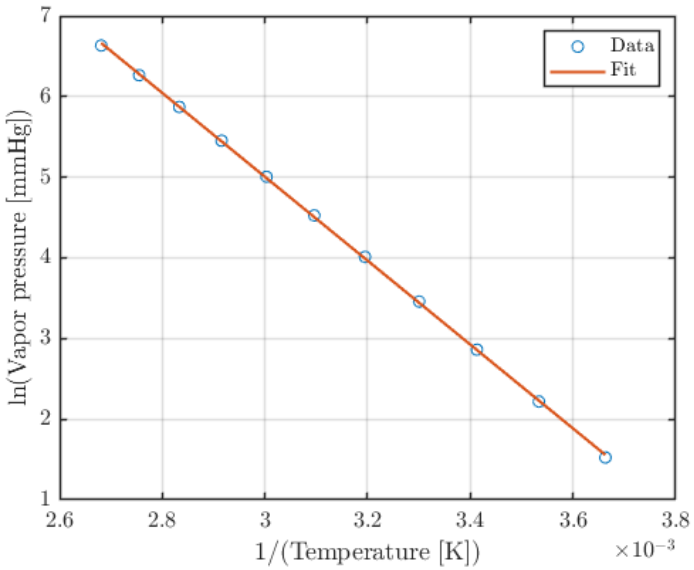


Figure 11: Plot of transformed data with fit.

Now we can test for significance of regression again with:

$$t_0 = \frac{-5200.8 - 0}{\sqrt{4.2735 \cdot 10^{-4}/(1.054 \cdot 10^{-6})}} = -258.2934$$

Again because $t_0 > t_{n-1,\frac{\alpha}{2}}$ we reject H_0 and say there is evidence to accept a linear relationship.

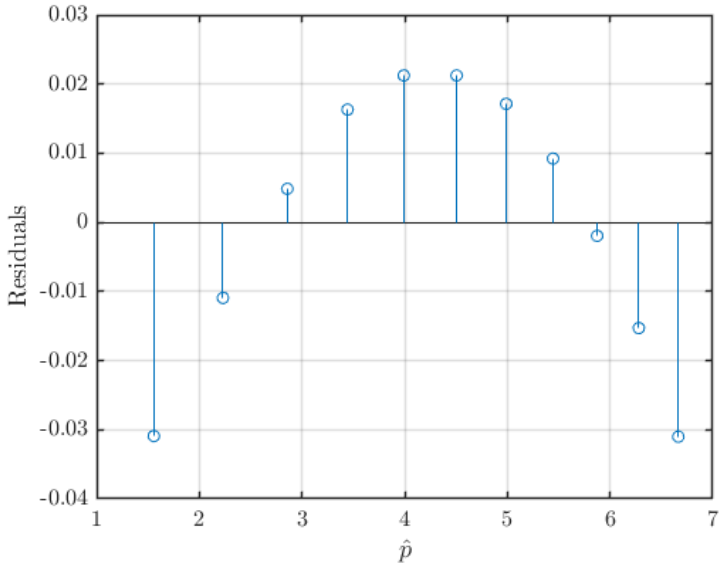


Figure 12: Plot of residuals.